



Lithospheric memory of subduction in mantle pyroxenite xenoliths from rift-related basalts

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ABSTRACT

Petrological and geochemical studies have revealed the contribution of garnet pyroxenites in basalt petrogenesis. However, whether primary mantle melts are produced with such signature or acquired it subsequently remains somewhat controversial. We here integrate new major-, trace-element and Sr-Nd-Hf isotopic compositions of garnet pyroxenite xenoliths in Holocene alkali basalts from Lakes Bullenmerri and Gnotuk, Southeastern Australia, to relate their petrogenesis to mantle-wedge melt circulation and subsequent lithospheric evolution. Results show that the clinopyroxenites have lower MgO and Cr₂O₃ contents than the associated websterites, and range in compositions from depleted LREE patterns and highly radiogenic Nd and Hf isotopic signatures in relatively low-MgO samples (Type 1), to enriched REE patterns with negative HFSE anomalies, unradiogenic Nd and Hf isotopes, and extremely radiogenic Sr-isotopic ratios in samples with higher MgO (Type 2). Such compositional variabilities suggest that these pyroxenites represent segregates from melts derived from a recycled oceanic lithosphere with a potential contribution from pelagic sediments. Variable LREE contents and isotopic compositions between those of Type 1 and 2 clinopyroxenites are observed in amphibole-bearing samples (Type 3), which are interpreted as Type 1-like protoliths metasomatized by the basaltic and carbonatitic melts, possibly parental to Type 2 clinopyroxenites. The lithosphere beneath Southeastern Australia thus has received variable melt contributions from a heterogeneous mantle-wedge source, which notably includes a subducted oceanic slab package that has retained its integrity during subduction. On this basis, we suggest that the compositional heterogeneity and temporal evolution of the subsequent Southeastern Australian basaltic magmatism were probably affected by the presence of pyroxenite fragments in the basalt source and formed by the tectonic reactivation of this lithosphere during Cenozoic rifting. This interpretation is notably consistent with a trend of Nd-Pb isotopes towards EMII in Older Volcanic Provinces (OVP basalts) and limited Sr-Nd-Pb isotopic variations towards HIMU in the Newer Volcanic Provinces (NVP basalts, including the host lavas), which also exhibit low SiO₂, high FeO and high CaO/Al₂O₃ commonly interpreted as due to pyroxenite contributions. Therefore, the identification of a subduction signature in these rift-related lavas attests to a “lithospheric memory” of earlier subduction episodes (as documented by the xenoliths), rather than a reflection of contemporaneous subduction tectonics.

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1. Introduction

Oceanic and continental crust may enter the mantle by subduction and delamination processes, potentially playing a fundamental role in the production of spatial heterogeneity and the temporal evolution of the Earth's mantle. Indeed, the isotopic and trace-element variability of ocean island basalts (OIB) and island arc basalts (IAB) demonstrates the presence of recycled crust in the mantle (Stracke, 2012; Hofmann, 2014). Minor amounts of crustal

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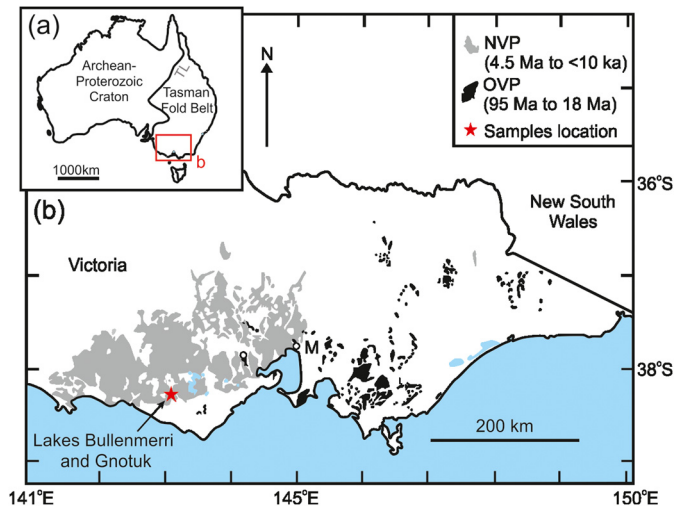


Fig. 1. (a) Map of the Australian plate; (b) Distribution of Older (95 Ma to 18 Ma) and Newer (<4.5 Ma) Volcanic Provinces (OVP and NVP, respectively) in Victoria and sample location, southeastern Australia. M, Melbourne.

components are also suspected in the source of mid-oceanic ridge basalts (MORB) (*i.e.* Hirschmann and Stolper, 1996). The involvement of such components in the petrogenesis of OIB and MORB has been interpreted as reflecting the presence of pyroxenite and/or eclogite in their source (*i.e.* Hirschmann and Stolper, 1996; Hirschmann et al., 2003). Experimental studies have confirmed that crustal signatures can be indeed acquired via partial melting of these mafic lithologies (*i.e.* Yaxley and Green, 1998; Hirschmann et al., 2003; Pertermann and Hirschmann, 2003; Kogiso and Hirahara, 2004). Moreover, pyroxenite layers within orogenic and ophiolitic peridotites provide direct evidence of this mantle heterogeneity; some of them exhibit variable isotopic compositions and positive Sr and Eu anomalies as expected from recycled lower oceanic crust in particular (*i.e.* Kornprobst et al., 1990; Pearson et al., 1993; Borghini et al., 2013; Le Roux et al., 2016).

Despite this overall agreement on the contribution of mafic lithologies in the source of mantle-derived magmas, the origin and evolution of recycled pyroxenites are still not fully understood (*i.e.* Kogiso and Hirahara, 2004; Stracke, 2012; Lambart et al., 2013). A number of processes can lead to the formation of pyroxenites in the mantle, including (1) recycling of subducted oceanic crust, stirred by mantle convection and incorporated into the lithosphere at different length scales (*i.e.* Allègre and Turcotte, 1986; Kornprobst et al., 1990), (2) high-pressure crystal segregation from mantle- and/or slab-derived melts (*i.e.* Pearson et al., 1993; Bizimis et al., 2005; Xiong et al., 2014), (3) melt-rock reaction between percolating melt and peridotites (*i.e.* Garrido and Bodinier, 1999; Liu et al., 2005; Borghini et al., 2016).

In Southeastern (SE) Australia, alkali basalts of the Newer Volcanic Province (NVP), which represents the most recent stage of volcanism (Fig. 1), have entrained abundant mantle-derived and crust-derived xenoliths. Among them, a suite of garnet pyroxenite xenoliths from maars at Lakes Bullenmerri and Gnotuk, exhibits an especially wide range of major- and trace-element and Sr- and Nd-isotope compositions (Griffin et al., 1984, 1988; Lu et al., 2017). Garnet websterites have high MgO contents (>17.1 wt%) and have been interpreted as cumulates from tholeiitic magmas, produced by hydrous melting of fertile mantle-wedge peridotites, previously metasomatized by fluids/melts from subducted materials (Lu et al., 2018). In contrast, garnet clinopyroxenites with low MgO contents (<17.1 wt%) show extreme Sr-Nd-Hf isotopic variations and inverse relationships between trace-element enrichment and fractional-crystallization indices (*e.g.* Mg#) that remain unresolved. We here use the major- and trace-element and Sr-Nd-

Hf isotope compositions of clinopyroxene (Cpx) and garnet (Grt) to explore the possibility that this difference reflects a heterogeneous mantle-wedge source and potential contributions of subducted oceanic lithosphere. The Grt pyroxenites record a lengthy decompressional path from their formation at ~75-km depth during Paleozoic subduction/accretion episodes, to re-equilibration at 40–60 km during Cenozoic rifting associated with the breakup of Gondwana (Lu et al., 2018). Based on these relationships between mantle processes, vertical tectonics mapped in the overlying crust, and regional volcanism, we show that the pyroxenites contribute to a “lithospheric memory”, passing on subduction fingerprints to subsequent rift-related volcanism.

2. Sample description and methods

Garnet clinopyroxenite xenoliths used for this study were collected from Holocene basanite tuffs (11.5–9.5 ka) in Lakes Bullenmerri and Gnotuk, which are roughly coeval vents, now maars, located <1 km apart (Fig. 1). Petrography and thermobarometry of the websterites have been investigated in detail to reconstruct the paleo-geotherm of southeastern Australia (SEA geotherm; Griffin et al., 1984; O'Reilly and Griffin, 1985) for 4-D lithosphere mapping (O'Reilly and Griffin, 1996). However, despite their abundance, clinopyroxenite xenoliths have only been the subject of limited work (Griffin et al., 1984, 1988) and little is known on their origin and the source of their parental magmas.

The Grt clinopyroxenite xenoliths analyzed in this study have round to angular shapes and are typically 5–10 cm in diameter (Fig. 2a–b). They show porphyroblastic to mosaic textures preserving macrocrysts and matrix minerals. The macrocrysts are mainly Cpx (up to 20 mm across) with exsolution lamellae of Grt ± plagioclase (Pl; Fig. 2c, d); matrix minerals are well recrystallized resulting in a fine-grained (0.5–2 mm) and equigranular mosaic texture of Cpx+Grt ± Pl ± Amphibole (Amp; Fig. 2e). In some samples, a transition from lamellar to mosaic texture is observed (Fig. 2d). In addition, some of the samples have coarse-grained spinel (Spl; up to 10 mm) commonly rimmed by Grt (Fig. 2f, g). They can be divided into three sub-types on the basis of mineral assemblages and mineral compositions (Table S1).

Type 1 samples are essentially bi-mineralic garnet clinopyroxenites with the exception of sample BM99-9, which contains minor Pl (Fig. 2c). Samples BM85-1 and GN41 are characterized by mosaic textures with coarse-grained Grt (2–5 mm) and Cpx (1–3 mm; Fig. 2a); samples BM99X2 and BM99-9 show porphyroblastic textures with most Grt and/or Pl occurring as exsolution lamellae within large Cpx macrocrysts (~6 mm; Fig. 2c, d), while rounded and/or polygonal neoblastic Grt (1–2 mm) locally preserve an alignment among randomly oriented Cpx subgrains and neoblasts (Fig. 2e). This observation indicates that they were probably exsolved lamellae from an originally coarse-grained Cpx that recrystallized to fine-grained neoblasts.

Type 2 samples are Spl-Grt clinopyroxenites composed of Cpx, Grt, Spl, Amp, and minor orthopyroxene (Opx), olivine (Ol) and apatite (Ap), and they show porphyroblastic textures with coarse-grained Spl and Cpx. The Spl grains are irregular to amoeboid with lobate outlines, commonly mantled by a granular assemblage of Grt and anhedral Ol (Fig. 2f, g); Ap and Amp also occur in the Spl-Grt reaction zones (Fig. 2g). The textural evidence suggests that the coarse Spl was a primary phase and Grt has grown through reaction between spinel and pyroxene and/or a metasomatic melt/fluid (Neal and Nixon, 1985). Rarely, Cpx occurs as macrocrysts with exsolution lamellae of Grt, Spl and/or Opx.

Type 3 samples are Grt clinopyroxenites with minor Amp, Ap and Pl and most have mosaic microstructures with coarse-grained Grt (3–4 mm) and Cpx (1–2 mm; Fig. 2h), while sample BM99X1 shows a porphyroblastic texture and is crosscut by Amp-bearing

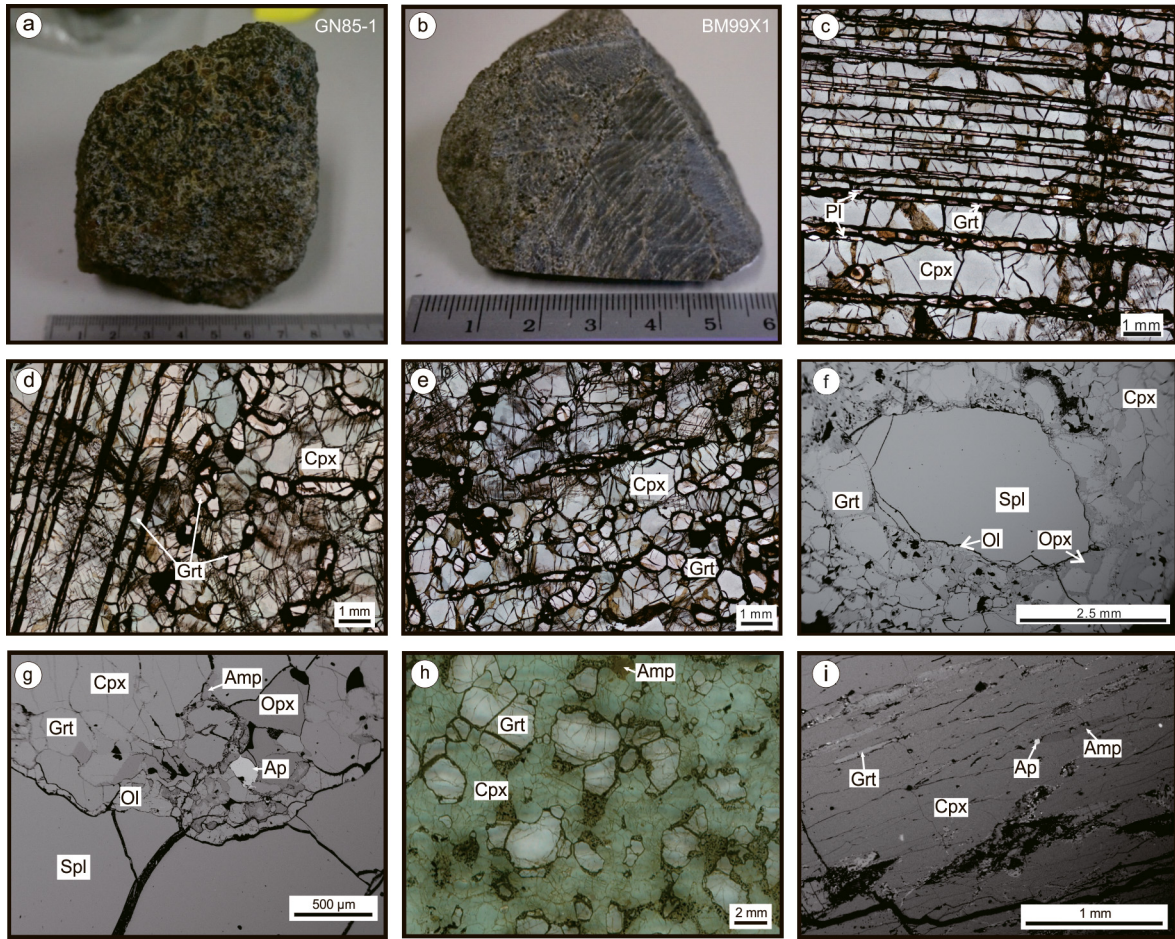


Fig. 2. Photomicrographs of Type 1 (a, c, d and e), 2 (f and g) and 3 (b, h and i) Grt clinopyroxenites. (a) Bimineralic Grt clinopyroxenite GN85-1, showing coarse-grained mosaic texture (hand specimen). (b) Garnet clinopyroxenite BM99X1, showing large Cpx with exsolution lamellae of Grt in lower part, and recrystallized, mosaic texture in upper part (hand specimen). (c) Clinopyroxene macrocryst with Grt and Pl lamellae. Plagioclase lamellae are always around Grt boundaries parallel to or crossing the Grt lamellae (PPL). (d) Transition from coarse-grained Cpx with exsolution lamellae of Grt, to mosaic texture of Cpx+Grt (BM99X2; PPL). (e) Equilibrated mosaic microstructure with lamellar residue of Grt. (f, g) Garnet corona around coarse Spl accompanied by Ol and clusters of small Spl, Opx, Amp and rare Ap (BSE). (h) Coarse texture showing coarse Grt surrounded by finer-grained Cpx (PPL). (i) Clinopyroxene macrocryst in sample BM99X1 with exsolution lamellae of Grt. Note that Amp and Ap locally occur as exsolution-like streaks (BSE). BSE, back-scattered electron; PPL, plane polarized light.

veins (Fig. 2b). Moreover, Amp-Ap exsolution-like streaks also occur on the recrystallized Cpx grains (Fig. 2i). No unambiguous contact relationships have been observed between the three pyroxenite types and the ambient peridotites. However, several Spl pyroxenite xenoliths, which were considered to have a similar origin and history to the Grt pyroxenites, occur as layers or pods in the host spinel lherzolites, suggesting that they are intrusive into the lherzolite (Griffin et al., 1984).

Major-element compositions of minerals were determined by electron microprobe analysis (EPMA) using a CAMECA SX100 instrument fitted with 5 wavelength dispersive spectrometers (Table S2). Trace-element compositions of silicates were measured by laser-ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) using an Agilent 7700 ICPMS coupled with a 193 nm ArF EXCIMER laser (Table S3). In order to assess both intra- and inter-grain chemical homogeneity, several grains per sample were analyzed for each phase, systematically testing for potential core-to-rim zoning. Bulk-rock major-element compositions were reconstructed using mineral analyses and point-counted modes (Table S4). Whole-rock compositions of several clinopyroxenite xenoliths determined by X-ray fluorescence (XRF) were taken from Griffin et al. (1988).

Sr- and Nd-isotopic ratios of separated Cpx, Amp and Grt were analyzed in solution using a Thermo Finnigan Triton thermal ion-

ization mass spectrometer (TIMS), and Hf-isotope ratios were determined in solution using a Nu Plasma multi-collector (MC)-ICP-MS (Table 1). Sr isotopes of Cpx, Amp, Ap and Pl were also analysed *in situ* using a 193 nm ArF EXCIMER laser system attached to Nu Plasma MC-ICPMS (Table S5). Detailed descriptions of the sample preparation and analytical procedures are given in the supplementary text.

3. Results

3.1. Major-element compositions

Clinopyroxenes from the clinopyroxenites have lower MgO, Cr₂O₃ and more variable Al₂O₃ and Na₂O contents than those of the websterites from the same locality (Fig. 3). Major-element compositions of Cpx in the clinopyroxenites also vary between subtypes and show marked negative correlations between Al₂O₃, Na₂O and MgO contents (Fig. 3). Specifically, Cpx in Type 1 clinopyroxenites has the lowest MgO (12.0–13.2 wt%) contents, but the highest Al₂O₃ (7.5–9.5 wt%) and Na₂O (1.6–2.0 wt%) contents. In contrast, Cpx in Type 2 clinopyroxenites has the highest MgO (14.6–15.3 wt%) but the lowest Na₂O (0.70–0.86 wt%) and Al₂O₃ (5.0–5.4 wt%) contents, plotting within the field of the websterites.

Table 1

Sr-Nd-Hf isotopes of clinopyroxene, amphibole, garnet and whole rock of garnet clinopyroxenites from Lakes Bullenmerri and Gnotuk, SE Australia.

Rock type	Sample No.	Mineral	Rb (ppm)	Sr (ppm)	⁸⁷ Rb/ ⁸⁶ Sr	⁸⁷ Sr/ ⁸⁶ Sr	2 SE	Sm (ppm)	Nd (ppm)	¹⁴⁷ Sm/ ¹⁴⁴ Nd	¹⁴³ Nd/ ¹⁴⁴ Nd	2 SE	Lu (ppm)	Hf (ppm)	¹⁷⁶ Lu/ ¹⁷⁷ Hf	¹⁷⁶ Hf/ ¹⁷⁷ Hf	2 SE
	BHVO-2					0.703468	0.000008				0.512985	0.000003				0.283107	0.000010
Type 1	BM99X2	Cpx	b.d.	13.9	-	0.704676	0.000009	0.64	0.58	0.6872	0.513547	0.000005	0.01	0.28	0.0025	0.283580	0.000012
		Grt	b.d.	0.08	-	-	-	0.36	0.07	3.0629	0.514189	0.000055	0.73	0.07	1.3751	0.286950	0.000015
		WR [#]			-	-	-	0.52	0.41	0.8149	0.513582	0.000006	0.13	0.20	0.0928	0.283801	0.000012
	BM85-1	Cpx	0.03	59.0	0.0012	0.704894	0.000002	0.82	2.77	0.1867	0.513557	0.000007	0.02	0.45	0.0061	0.283963	0.000011
		Grt	0.04	0.47	0.2402	-	-	0.46	0.21	1.3935	0.513997	0.000035	1.29	0.11	1.6177	0.289316	0.000006
		WR [#]			-	-	-	0.62	1.32	0.2958	0.513602	0.000010	0.73	0.27	0.3792	0.285409	0.000011
	BM99-9	Cpx	0.02	7.93	0.0068	0.704924	0.000009	0.23	0.77	0.1881	0.512964	0.000030	0.01	0.11	0.0158	0.283479	0.000042
	GN41 ⁺	WR	0.30	33.1	0.0256	0.704590	0.000010	0.85	1.51	0.3406	0.513129	0.000010	-	-	-	-	-
	Type 2	BM90-3	Cpx	0.14	127	0.0031	0.715485	0.000004	7.41	40.8	0.1143	0.512221	0.000006	0.05	1.39	0.0050	0.282747
BMC-1		Cpx	b.d.	138	-	0.714232	0.000004	7.14	40.0	0.1126	0.512218	0.000002	0.01	1.10	0.0018	0.282774	0.000011
		Amp	3.80	580	0.0185	0.715119	0.000004	11.7	60.5	0.1214	0.512216	0.000001	0.04	0.99	0.0058	0.282791	0.000005
DR11022 ⁺		WR	0.24	121	0.0056	0.715756	0.000014	6.47	32.4	0.1213	0.512201	0.000014	-	-	-	-	-
DR11017c ⁺		WR	0.63	133	0.0134	0.714655	0.000012	6.60	32.0	0.1240	0.512235	0.000008	-	-	-	-	-
DR11017a ⁺		WR	0.62	145	0.0121	0.714838	0.000010	6.49	32.2	0.1223	0.512222	0.000018	-	-	-	-	-
Type 3	BM99X1	Cpx	0.12	120	0.0027	0.709112	0.000009	1.95	8.46	0.1449	0.512424	0.000003	0.03	1.07	0.0035	0.282837	0.000004
	GN18 ⁺	WR	0.29	100	0.0010	0.704948	0.000010	1.41	7.10	0.1252	0.512565	0.000013	-	-	-	-	-
	DR9734 ⁺	WR	-	-	-	0.703732	0.000016	-	-	-	0.512858	0.000010	-	-	-	-	-

b.d. Below detection.

^{*} Data from Griffin et al. (1988).[#] Whole-rock Nd-Hf isotopic compositions were reconstructed from Cpx and Grt, which dominate the Sm, Nd, Lu and Hf budget.

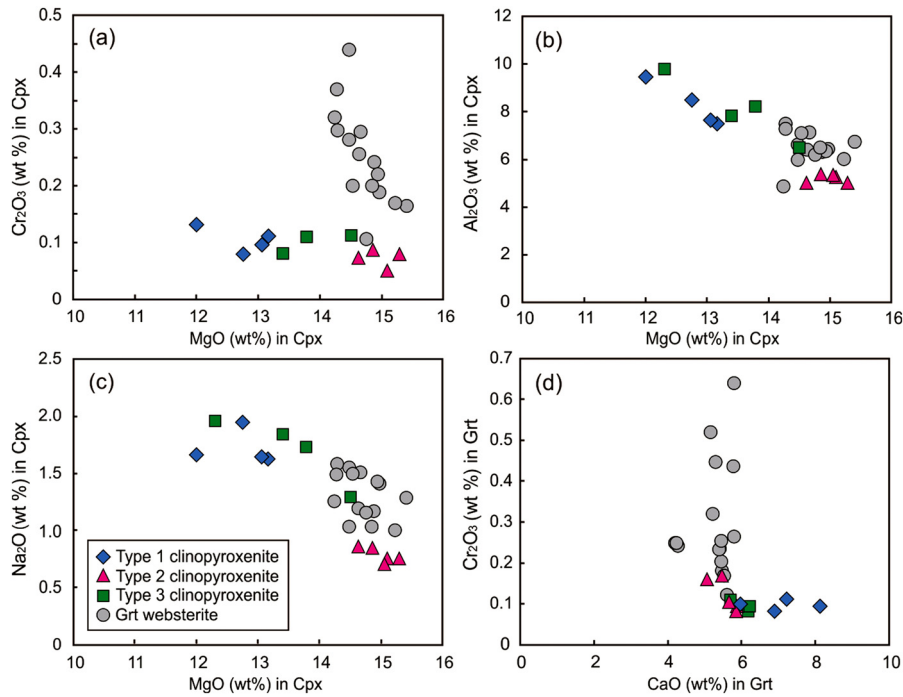


Fig. 3. Major-element compositions of Cpx (a, b, c) and Grt (d) in Grt pyroxenites from Lakes Bullenmerri and Gnotuk, SE Australia.

Cpx in Type 3 clinopyroxenites has compositions intermediate between the other two sub-types for most oxides.

Garnet from the clinopyroxenites has lower MgO and Cr_2O_3 but higher CaO contents than those of the websterites (Fig. 3), and also varies between the subtypes. Specifically, Grt in Type 1 Grt clinopyroxenites has relatively high but variable CaO (6.0–8.1 wt%) and low Cr_2O_3 (mostly <0.1 wt%). Garnet in Type 2 clinopyroxenites shows intermediate MgO (and Mg#) and low CaO and Cr_2O_3 contents, plotting at the intersection between the trends defined by the websterites and Type 1 clinopyroxenites. Grt in Type 3 clinopyroxenites again has compositions intermediate between the other two sub-types.

Amphibole and apatite mainly occur in Type 2 clinopyroxenites but minor amounts are also found in Type 3 clinopyroxenites. Amp is pargasitic and has lower Cr_2O_3 than that in the websterites (Table S2). Type 2 Amp shows lower TiO_2 , K_2O but higher F contents than those of Type 3. Ap in Type 2 clinopyroxenites has high F (1.1–1.5 wt%) and low Cl contents (0.20–0.89 wt%), corresponding to the cumulus Ap defined by O'Reilly and Griffin (2000). In contrast, Ap in a Type 3 clinopyroxenite (BM99X1) is hydroxylchlorapatite with low F (0.63 wt%) and high Cl (2.2 wt%) contents and probably is of metasomatic origin (Table S2).

Major-element compositional profiles indicate that constituent minerals are homogeneous within individual samples (Lu et al., 2017). There is no compositional gradient between the host Cpx and exsolved minerals, suggesting that the garnet pyroxenite xenoliths reached chemical equilibrium before entrainment in the host basalt (Table S2; Lu et al., 2017). Bulk-rock compositions were thus reconstructed using the average compositions of the main rock-forming minerals (Cpx, Grt, Opx, Amp and Spl) and their modes determined by point counting (Table S4). Comparison between our reconstructed bulk-rock compositions and whole-rock compositions of some of the samples analysed using X-ray fluorescence (XRF) by Griffin et al. (1988) shows a good agreement in terms of major elements, indicating the reliability of our reconstructions. They exhibit low MgO (Mg#), Cr_2O_3 contents (Fig. 4a) and variable CaO and Al_2O_3 contents (Fig. 4b). Mg# is positively correlated to CaO and SiO_2 (Fig. 4b, 4c) and negatively correlated to Al_2O_3 ;

the latter also decreases with increasing CaO (Fig. 4d). The clinopyroxenites do not show a compositional continuum with the websterites, which plot in the field of experimental silica-depleted (SD) pyroxenite residues (Fig. 4), but within the field of experimental residues of a single clinopyroxenite M7-16 from Beni Bousera.

3.2. Trace-element compositions

The three sub-types of Grt clinopyroxenites are further distinguished by their trace-element compositions. In particular, Cpx from most Type 1 clinopyroxenites have relatively depleted LREE and fractionated MREE-HREE [(Sm/Yb)_N = 3.97–20.6; N indicates normalization to chondrites; McDonough and Sun, 1995], although they display variable enrichment in La and Ce (Fig. 5a). They also show slightly positive Eu, Pb, Sr and Ti anomalies (Fig. 5b) and fractionated Zr/Hf ratios [(Zr/Hf)_{PM} = 0.25–0.65; PM indicates normalization to the Primitive Mantle; McDonough and Sun, 1995]. Cpx from Type 2 clinopyroxenites have convex-upward REE patterns [(La/Nd)_N = 0.57–0.73; (Sm/Yb)_N = 31.7–56.7] coupled with enrichment in large-ion lithophile elements (LILE) and depletion in high field-strength elements (HFSE) (Fig. 5c, d). They also have negative Eu [(Eu/Eu*)_N = 0.81–0.83], Pb and Sr anomalies, and fractionated Zr/Hf ratios [(Zr/Hf)_{PM} = 0.53–0.59]. Cpx from Type 3 pyroxenites display trace-element patterns intermediate between those of Type 1 and Type 2, with LREE enrichment [(La/Nd)_N = 1.1–2.4] and negative Pb and HFSE anomalies, but no clear Eu or Sr anomalies (Fig. 5e, f). The clinopyroxenitic Cpx are thus different from those of the websterites, which are characterized by slightly enriched LREE [(La/Nd)_N = 0.77–2.22], positive Eu [(Eu/Eu*)_N = 1.0–1.4] and Sr anomalies, and the absence of a Ti anomaly.

Garnet also differs markedly between clinopyroxenite sub-types (Fig. 5g; Table S3). Specifically, Type 1 Grt has the lowest LREE contents and high HREE contents [(Yb/Gd)_N = 2.5–4.6], except for lamellar and granular Grt in BM99X2 which have negative HREE slopes. In contrast, Type 2 Grt have the highest bulk REE contents and strong LREE-to-MREE fractionation [(Sm/La)_N = 132–310] but only slightly positive MREE-to-HREE slopes [(Yb/Gd)_N = 1.4–3.0].

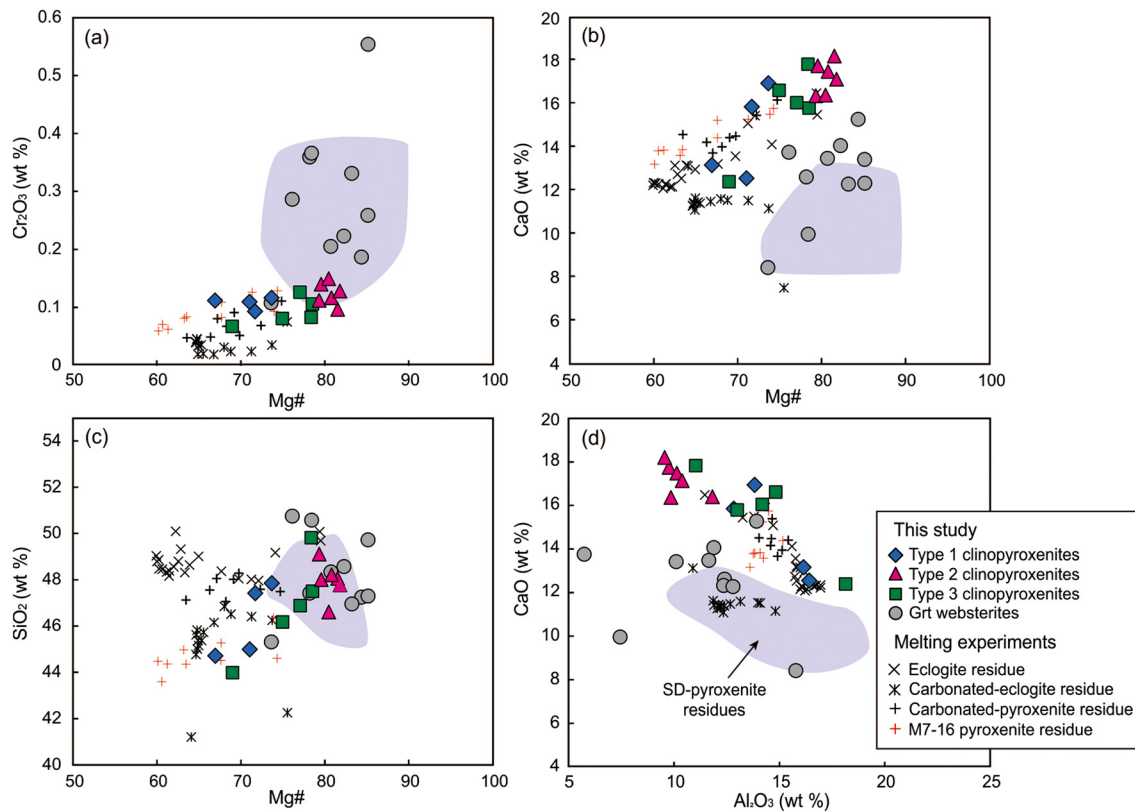


Fig. 4. Whole-rock major oxides vs Mg# (a, b, c) and Al_2O_3 vs CaO (d) for garnet pyroxenites from Lakes Bullenmerri and Gnotuk, SE Australia. Experimental solidus residues from MORB-like eclogite (cross; Pertermann and Hirschmann, 2003), carbonated eclogite (asterisk; Dasgupta et al., 2006), carbonated MORB-like pyroxenite (black plus sign; Gerbode and Dasgupta, 2010), silicate-deficient (SD) pyroxenite (purple shading; Keshav et al., 2004; Lambart et al., 2013) and a single clinopyroxenite M7-16 from Beni Bousera with lower MgO contents than SD pyroxenites (red plus sign; Lambart et al., 2013) at 2–3 GPa are shown for comparison. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

Type 3 Grt have LREE intermediate between the other two types, but mostly flatter MREE-to-HREE slopes.

Amphibole shows REE patterns similar to those of Cpx, only differing in their slightly higher ΣREE , and higher LILE (Rb, Ba and K) and Nb contents, and positive Nb/Ta [$(\text{Nb}/\text{Ta})_{\text{PM}} = 1.2\text{--}4.7$] (Table S3). Trace-element patterns of Ap also mimic those of Cpx but higher contents, with the exception of one Ap grain coexisting with Spl and Grt in a Type 2 clinopyroxenite (BMC-1; Fig. 2g), which exhibits extremely depleted HREE compared to Ap included in Cpx (Fig. 5h). This observation suggests that this Ap crystallized coevally with the Spl–Grt re-equilibration, in good agreement with its lower Sc, Zn and Al contents (Table S3).

In most samples, no core-to-rim compositional gradients have been observed in Grt and Cpx within individual samples. The exception is Grt in BM99X2 where there is a marked difference in HREE and Y contents between the rare coarse Grt and lamellar Grt (Lu et al., 2017). The trace-element compositions of bulk rocks reconstructed using the average compositions of Cpx, Grt, Amp and Opx show distinct LREE patterns between sub-types, as for Cpx (Table S4). Considering that Grt hosts little of the highly incompatible trace elements, the LILE and LREE budgets of the bulk rock are mainly controlled by Cpx, but Grt expectedly dominates the HREE budget. Type 1 and 3 clinopyroxenites have flat HREE-rich patterns contrasting with those of Type 2 clinopyroxenites (Table S4).

3.3. Pressure and temperature estimates

Three clinopyroxenites containing Cpx macrocrysts were reconstructed to estimate their initial crystallization conditions (Lu et al.,

2017). Using thermodynamic modelling in the system $\text{CaO-MgO-Al}_2\text{O}_3\text{-SiO}_2$ (CMAS; Gasparik, 2014), Type 1 samples yield slightly higher P–T estimates (1400 °C and 1.7–2.0 GPa) than Type 3 samples (1280 °C and 1.6 GPa; Table S1), both being lower than those obtained from the websterites (1420–1450 °C and 2.3–2.4 GPa; Lu et al., 2018).

For Opx-bearing Type 2 samples, P–T estimates based on the Cpx–Grt Fe–Mg exchange thermometer of Ellis and Green (1979) and Al-in-Opx barometer of Wood (1974) were also calculated, yielding 950–1005 °C and 1.2–1.5 GPa (Table S1). For Opx-free xenoliths, the equilibrium pressure was estimated from the intersection between temperatures obtained with the Cpx–Grt thermometer and the SEA geotherm (O'Reilly and Griffin, 1996), yielding 1.3–1.8 GPa at 1000–1090 °C for Type 1 samples and 1.2–1.8 GPa at 970–1100 °C for Type 3 samples (Table S1). These results are comparable to those obtained with the REE-in-Grt–Cpx thermobarometer (Sun and Liang, 2015) for exsolved Grt and host Cpx. In addition, coarse Grt and adjacent Cpx in BM99X2 yield intermediate P–T conditions (1032 °C, 2.1 GPa), indicating that these pyroxenites followed a decompressional cooling path before entrainment in the host basalt (Lu et al., 2017).

3.4. Sr-, Nd-, and Hf-isotope geochemistry

The Rb–Sr, Sm–Nd and Lu–Hf isotopic compositions of pyroxenitic Grt, Cpx and Amp, together with the published data of Griffin et al. (1988), are given in Table 1. We observe that the xenolith types identified on the basis of major- and trace-element compositions also have distinct isotopic signatures (Fig. 6). Specifically, Cpx from Type 1 clinopyroxenites have relatively homogeneous $^{87}\text{Sr}/^{86}\text{Sr}$ (0.7046–0.7049) and highly radiogenic $^{143}\text{Nd}/^{144}\text{Nd}$

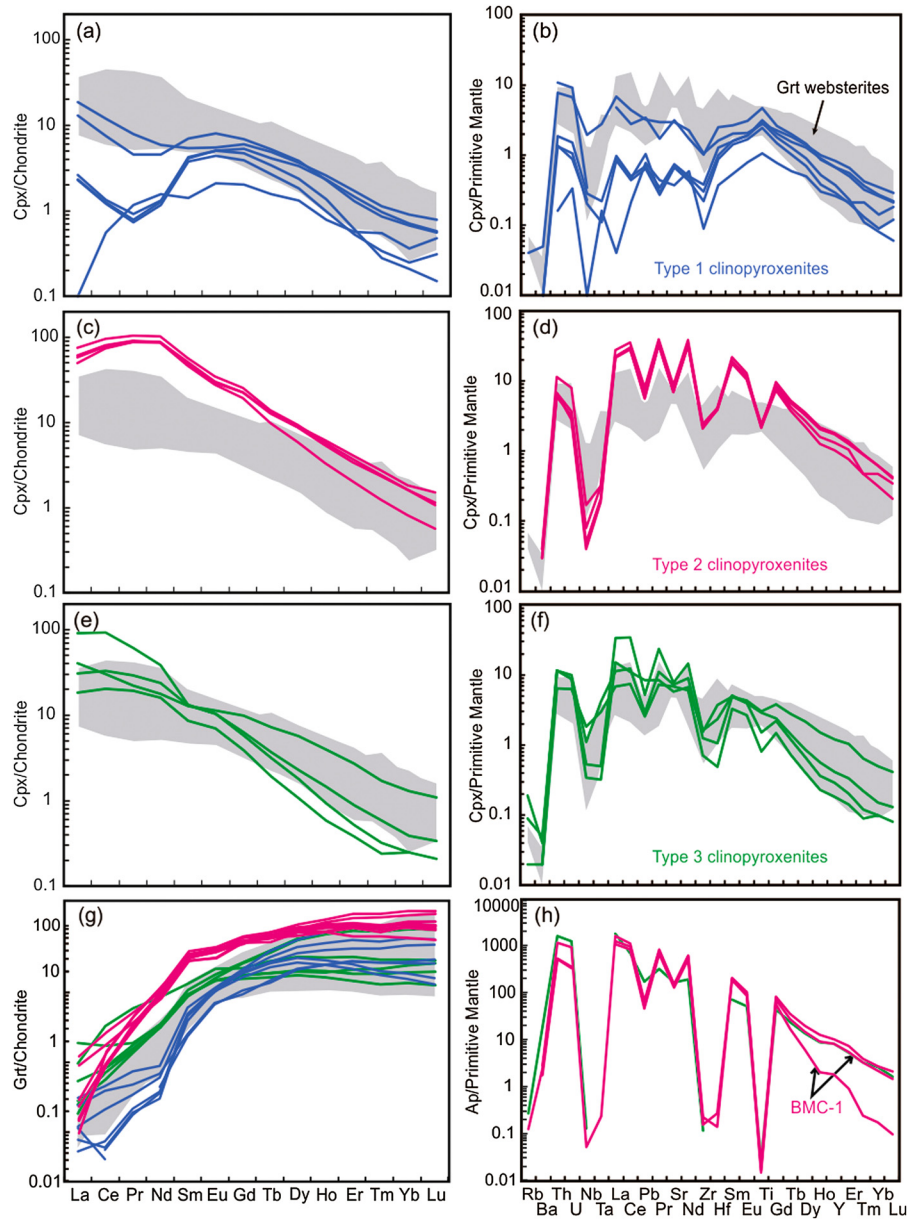


Fig. 5. Chondrite-normalized and primitive mantle-normalized trace-element patterns of clinopyroxene (a–f), garnet (g) and apatite (h) in garnet pyroxenite xenoliths from Lakes Bullenmerri and Gnotuk, SE Australia. Chondrite and primitive-mantle values are from McDonough and Sun (1995). Shaded areas represent garnet websterite xenoliths from the same area.

(0.5129–0.5136; Fig. 6a), especially for samples with low Mg# (BM99X2 and GN85-1), which also have low Sr contents (13.9–59 ppm) and Nd/Sr (0.04–0.05; Fig. 6b). These compositions are shifted to higher $^{87}\text{Sr}/^{86}\text{Sr}$ from the MORB field in Sr–Nd space (Fig. 6a). In contrast, Cpx from Type 2 clinopyroxenites have homogeneous and unradiogenic $^{143}\text{Nd}/^{144}\text{Nd}$ (~ 0.5122) and radiogenic $^{87}\text{Sr}/^{86}\text{Sr}$ (0.7142–0.7157), high Sr contents (127–138 ppm) and Nd/Sr (0.29–0.32). Cpx from Type 3 clinopyroxenites have intermediate Sr- and Nd-isotope compositions, with high Sr (~ 120 ppm) comparable to that of Type 2, but low Nd/Sr as measured in Type 1 samples. In comparison, garnet websterites ($^{87}\text{Sr}/^{86}\text{Sr} < 0.7060$; $^{143}\text{Nd}/^{144}\text{Nd} > 0.5124$) overlap the field of the host peridotites within the MORB–OIB mantle array (Griffin et al., 1988; Stolz and Davies, 1988). *In-situ* Sr-isotope analyses of inclusion-free Cpx, Amp, and Pl are identical within error to those obtained by isotope dilution on Cpx and Amp separates (Fig. 6b; Table S5).

The Hf-isotope compositions of Cpx are positively correlated with their $^{143}\text{Nd}/^{144}\text{Nd}$ but plot slightly off the Nd–Hf mantle array towards more radiogenic Hf (Fig. 6c). In detail, Cpx from Type 1 clinopyroxenites have the most radiogenic Hf ($^{176}\text{Hf}/^{177}\text{Hf} = 0.2834$ – 0.2839) with relatively high Hf/Nd (0.14–0.47), while those from Type 2 pyroxenites have the least radiogenic Hf isotopes (0.2827–0.2828) and the lowest Hf/Nd (~ 0.03 ; Fig. 6d). Cpx from Type 3 pyroxenites and the websterites have intermediate compositions (Fig. 6c).

Amphibole has Nd–Hf isotopic compositions similar to those of coexisting Cpx but slightly more radiogenic Sr, especially in sample BMC-1 where Amp has $^{87}\text{Sr}/^{86}\text{Sr}$ higher than that of Cpx by $\sim 10^{-3}$ (Tables 1 and S5). Garnet has more radiogenic Nd and Hf than coexisting Cpx (Table 1). One Cpx–Grt pair from Type 1 clinopyroxenites gives Sm–Nd isochron age of 56 ± 16 Ma, and older Lu–Hf isochron age of 178 ± 100 Ma (not shown).

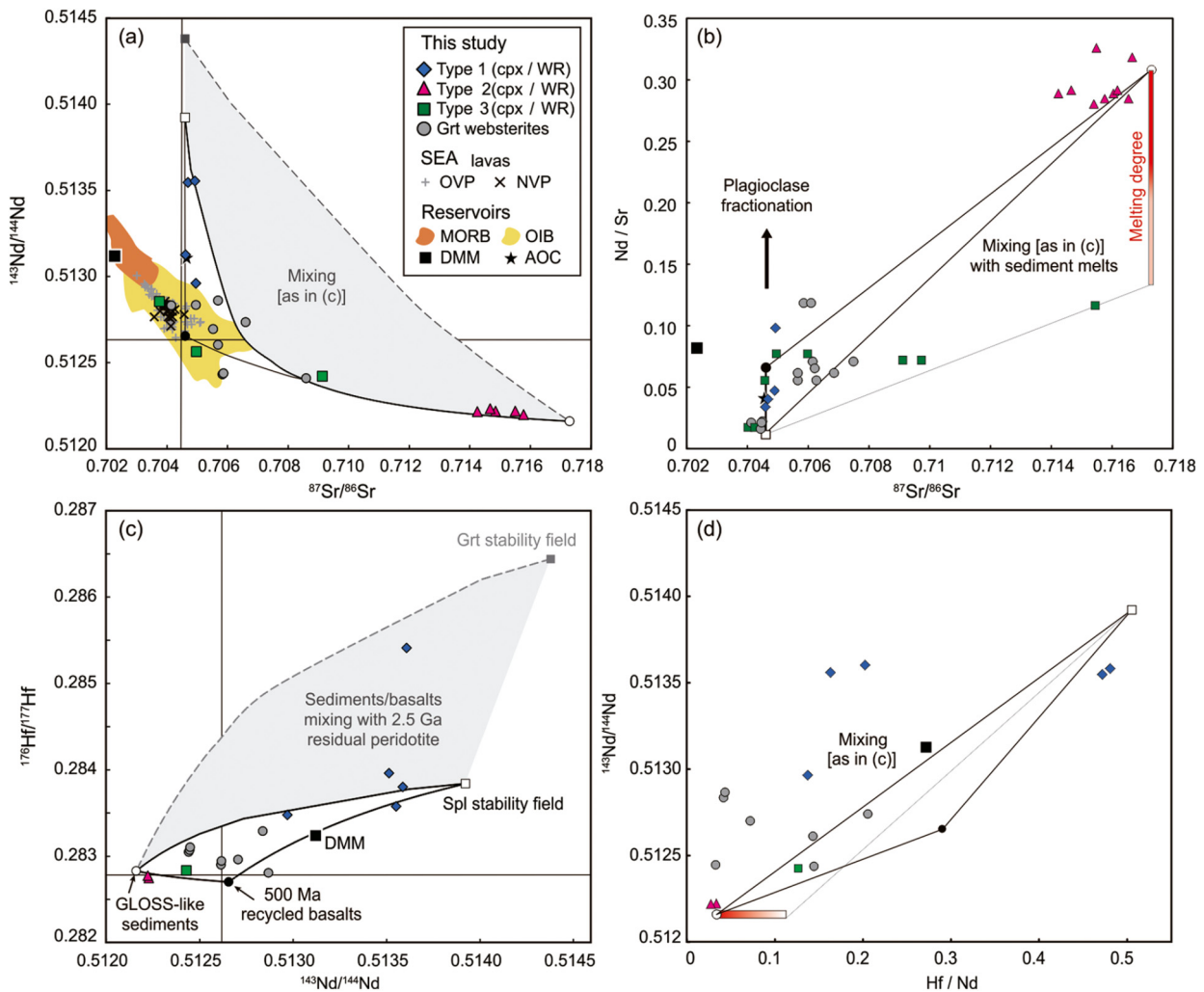


Fig. 6. Sr-Nd-Hf isotopic variations for garnet pyroxenites from Lakes Bullenmerri and Gnotuk, SE Australia. These pyroxenites mainly plot within the field defined by mixing of a metasomatized 2.5-Ga depleted peridotite, GLOSS-like sediments and a recycled basalt. Model compositions were obtained as follows. 2.5 Ga depleted peridotite: present-day composition of residue of 5% MORB melting at 2.5 Ga ago with $^{87}\text{Sr}/^{86}\text{Sr} = 0.7046$, $^{143}\text{Nd}/^{144}\text{Nd} = 0.5139$, $^{176}\text{Hf}/^{177}\text{Hf} = 0.28384$, $C_{\text{Sr}} = 10$ ppm, $C_{\text{Nd}} = 0.11$, $C_{\text{Hf}} = 0.06$. GLOSS-like bulk sediment: $^{87}\text{Sr}/^{86}\text{Sr} = 0.7073$, $^{143}\text{Nd}/^{144}\text{Nd} = 0.51216$, $^{176}\text{Hf}/^{177}\text{Hf} = 0.28283$, $C_{\text{Sr}} = 133$ ppm, $C_{\text{Nd}} = 41$ ppm, $C_{\text{Hf}} = 1.39$ ppm. Recycled basalt: 1% batch melting from the ancient mantle peridotite at 500 Ma with $^{87}\text{Sr}/^{86}\text{Sr} = 0.7046$, $^{143}\text{Nd}/^{144}\text{Nd} = 0.51265$, $^{176}\text{Hf}/^{177}\text{Hf} = 0.28270$, $C_{\text{Sr}} = 41.7$ ppm, $C_{\text{Nd}} = 2.75$ ppm and $C_{\text{Hf}} = 0.8$ ppm. The grey shaded areas (a, c) indicate the mantle end-member assuming partial melting in the garnet stability field. The average compositions of depleted MORB mantle (DMM) and altered oceanic crust (AOC) are from Workman and Hart (2005) and Nielsen and Marschall (2017), respectively. Additional mixing lines between the mantle and sediment melts (20% partial melting) are also shown in b and d (dashed lines). Partition coefficients for Sr, Nd and Hf during sediment melting were set to $D_{\text{Sr}} = 2$, $D_{\text{Nd}} = 0.35$ and $D_{\text{Hf}} = 0.7$, respectively, which represent the average values recorded in sediment melting experiments by Hermann and Rubatto (2009). Fields of MORB, OIB in (a) are from Vervoort et al. (1999). Data of OVP and NVP are from Price et al. (2014) and Oostingh et al. (2016), respectively.

The reconstructed bulk-rock compositions expectedly are more radiogenic in Nd and Hf than Cpx, but comparable in terms of Sr (Table 1).

4. Discussion

Among the Bullenmerri and Gnotuk xenoliths, the Grt websterites have been interpreted as cumulates of mantle-wedge melts derived from flux melting of a peridotite source metasomatized by slab-derived fluids or melts (Lu et al., 2018). The Grt clinopyroxenites studied here show microstructures related to exsolution and recrystallisation comparable to those of the websterites, and they have lower MgO and Ni contents, precluding the presence of Ol on the liquidus; olivine would be the main crystallizing phase in these mafic magmas at pressures <1.8 GPa (Bultitude and Green, 1971). This probably indicates that the clinopyroxenites also initially formed as Cpx ($\pm$ Spl) segregations at high pressures (Table S1). CaO is positively correlated with Mg#, while Al_2O_3 and Na_2O exhibit the opposite trend, which can be reasonably ascribed

to various degrees of fractional crystallization (Fig. 4). This might suggest that a more evolved magma, potentially residual after the crystallization of the websterites, was involved. However, trace-element and Sr-Nd-Hf isotopic compositions of the clinopyroxenites and websterites are inconsistent with such cogenetic relationships (Figs. 5 and 6). In addition, crystallization indexes (e.g. MgO, Yb) and the concentration of highly incompatible elements (i.e. Sr, LREE, Th) also argue against cogenetic relationships among the clinopyroxenites and rather point towards different melt sources and/or some extent of metasomatic overprint. Specifically, the most evolved pyroxenites (i.e. Type 1, with the lowest MgO and Mg#) also have the lowest abundances of incompatible elements, while the relatively MgO-rich (Type 2) clinopyroxenites have the most enriched REE patterns (Fig. 5). Instead, we demonstrate below that the compositional range observed among clinopyroxenites relates to cumulate (and interaction) processes involving melts with different geochemical affinities. We thus consider this alternative below with a particular emphasis on the potential involvement of subducted components.

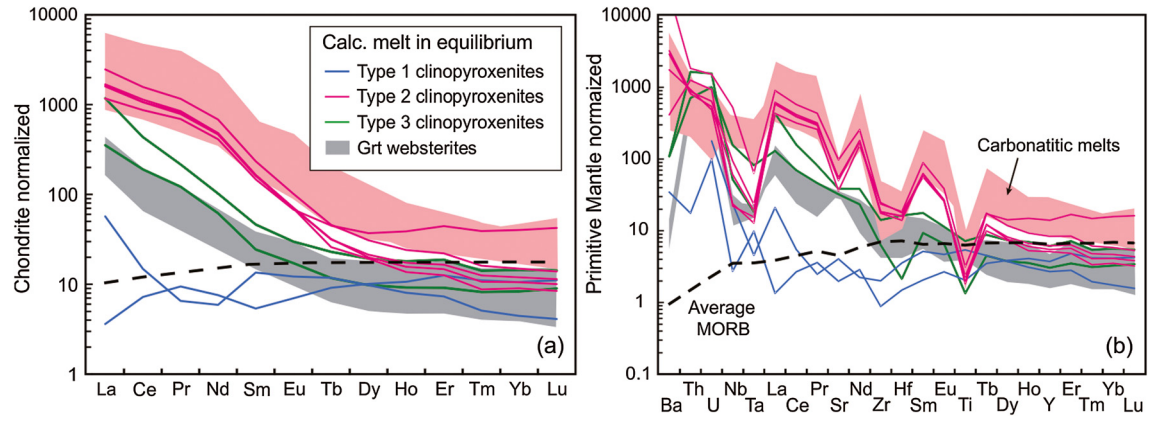


Fig. 7. Normalized REE patterns and spider diagrams for melts in equilibrium with garnet clinopyroxenites studied here from Lakes Bullenmerri and Gnotuk. The patterns were calculated using the reconstructed Cpx and mineral-melt partition coefficients listed by Adam and Green (2006) and Hart and Dunn (1993), except for Type 2 clinopyroxenites in which bulk-rock compositions are used for representing the reconstructed clinopyroxene. Shaded areas represent carbonatite melts (pink; Tappe et al., 2006) and garnet websterite xenoliths from the same area (grey; Lu et al., 2018), respectively; the data of average MORB (dash line) is from McDonough and Sun (1995).

4.1. An olivine-poor source from recycled oceanic crust

The computed melt compositions in equilibrium with primary Type 1 Cpx (Fig. 7) are closely similar to depleted MORB-like melts obtained after 5% fractional melting of the depleted mantle (DM) in the stability field of Grt peridotite (i.e., BM99X2) or 10% partial melting of Spl peridotite (i.e., BM99-9). However, these melts have low Mg# (~45) and SiO₂ but high Al₂O₃ contents (Table S6), which are different from the evolution of MORB and thus argues against their direct derivation from a peridotitic source. An olivine-poor (e.g. pyroxenitic or eclogitic) source with MORB-like trace-element characteristics would reconcile the depleted REE patterns and relatively low Mg#. Such source components have been proposed previously for LREE-depleted pyroxenites from orogenic and ophiolitic massifs, such as Ronda, Beni Bousera and the Bohemian Massif (i.e. Pearson et al., 1993; Garrido and Bodinier, 1999; Montanini et al., 2012; Borghini et al., 2016).

To better understand the source of Type 1 clinopyroxenites, experimental solidus residues on MORB-like eclogite, carbonated eclogite, carbonated MORB-like pyroxenite, silicate-deficient (SD) pyroxenite and a single clinopyroxenite M7-16 from Beni Bousera with lower MgO content than SD pyroxenites at 2-3 GPa were considered for comparison (Figs. 4 and 8). The compositions of Type 1 pyroxenites trend towards the partial-melting residues of eclogitized basalt at 2-3 GPa but they always have lower SiO₂ and Na₂O contents (Fig. 4). In addition, eclogitic residues always have silica-excess (SE) compositions, plotting on the right-hand side of the enstatite (En)-Ca-Tschermak (CaTs) join, along the forsterite (Fo)-anorthite (An) join (Pertermann and Hirschmann, 2003; Yaxley and Sobolev, 2007; Rosenthal et al., 2014, 2018) in the pseudoternary Fo-CaTs-quartz (Qz) system projected from diopside (Di) (Fig. 8). Type 1 clinopyroxenites plot along the Fo-An join but on the silica-deficient (SD) side of the En-CaTs join (Fig. 8), which argues against a purely eclogitic source. Experiments show that the presence of CO₂ in carbonated eclogites and pyroxenites (Dasgupta et al., 2006; Gerbode and Dasgupta, 2010) significantly decreases the SiO₂ and increases the MgO contents of the derived melts. The residues after partial melting of carbonated eclogite have low SiO₂ and Mg#, similar to Type 1 clinopyroxenites at 2-3 GPa (Dasgupta et al., 2006), but their carbonatitic liquids or carbonated silicate partial melts (T > 1200 °C) result in lower Ca contents in the residues than seen in Type 1 clinopyroxenites (Fig. 4). Moreover, experimental melting residues of carbonated MORB-like pyroxenite (Gerbode and Dasgupta, 2010) have compositions better approximating the Type 1 clinopyroxenites but higher Na₂O contents and trend towards CaTs in the Fo-CaTs-Qz [+Di] (Figs. 4 and 8).

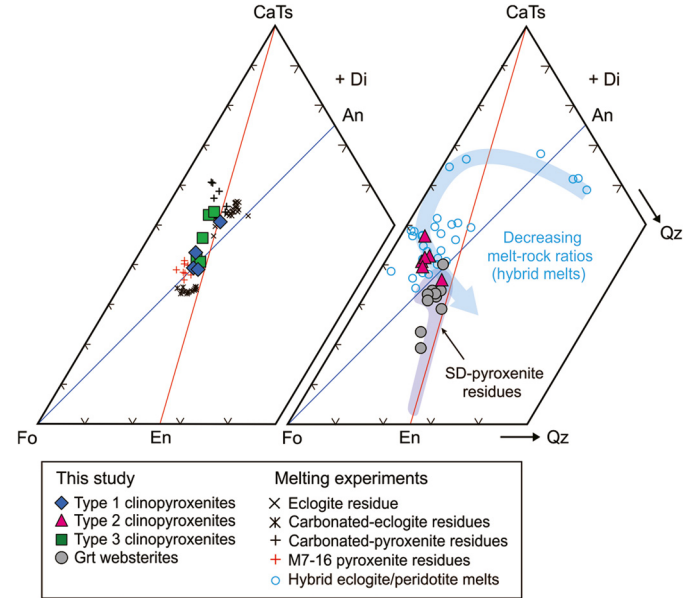


Fig. 8. Pseudoternary system forsterite (Fo)-calcium tschermak (CaTs)-quartz (Qz) projected from diopside (Di) comparing pyroxenite xenoliths from Lakes Bullenmerri and Gnotuk with experimental partial residues of M7-16 clinopyroxenite (red plus), MORB-eclogite (cross), carbonated eclogite (asterisk), carbonated MORB-like pyroxenite (black plus sign), high-MgO SD-pyroxenite (purple shaded area) and melts from eclogite-peridotite hybrids (circle) at 2-3 GPa. Data for the experimental melts are from (Kogiso et al., 1998; Yaxley and Green, 1998; Mallik and Dasgupta, 2012), and data for the experimental residues are the same as in Fig. 4. The arrow in b refers to the direction of melt-rock reaction with decreasing melt-rock ratio for mixed runs (Mallik and Dasgupta, 2012). In this system, nepheline- and hypersthene-normative compositions are separated by the Di-Fo-An plane (blue line). Silica-deficient and silica-excess compositions are divided by the Di-CaTs-En plane (red line).

Nonetheless, experiment on an olivine-poor pyroxenites suggests that Na₂O contents decrease with increasing melt fraction (higher potential temperature; i.e. Borghini et al., 2017), which could indicate that Type 1 clinopyroxenites presumably resulted by a melt produced by residual MORB-like eclogite/pyroxenite at deep hot mantle. Specifically, a clinopyroxenite (e.g. M7-16; geochemically comparable to carbonated pyroxenite with lower Na₂O content; Lambart et al., 2013) could account for the compositional discrepancy and yielded residual solid compositions nearly identical to those of Type 1 clinopyroxenites at high pressure (2-3 GPa) (Figs. 4 and 8). The latter plot on the trend line between these residues

and their complementary melts on Mg# vs CaO and SiO₂ diagrams (Fig. 4). In comparison, the residues derived from an even MgO-richer pyroxenites (*SD pyroxenites*) exhibit lower CaO and Al₂O₃ at a given MgO. We suggest that Type 1 clinopyroxenites may be segregates from magmas derived from a MORB-like (*i.e.* LREE-depleted) pyroxenite-rich source with a composition resembling natural pyroxenite M7-16 (Fig. 8). M7-16 is a nepheline-normative Grt-bearing clinopyroxenite, similar to the graphite-bearing Grt pyroxenites with relatively low SiO₂ (43.4 wt%) and low Mg# (60.6) from the Beni Bousera Peridotite (Morocco). They are interpreted as partially molten high-pressure crystal segregates from melts of hydrothermally altered oceanic crust (*i.e.* Pearson et al., 1993). It is therefore likely that low-Si, low-Mg pyroxenites dominated the source of the parental melts of Type 1 pyroxenites, potentially as a result of oceanic crust recycling.

This is consistent with their Nd-isotope compositions, which are comparable to that of MORB, while their more radiogenic Sr could be ascribed to hydrothermal alteration or fluid metasomatism (*i.e.* with high Sr/Nd; Kessel et al., 2005). However, some samples have Nd-Hf even more radiogenic than the DMM and show significant decoupling of Hf from Nd isotopes towards extremely radiogenic Hf compositions at nearly constant ¹⁴³Nd/¹⁴⁴Nd in Nd-Hf isotope space (Fig. 6c), probably suggesting the involvement of an ancient depleted component in their source (*i.e.* Bizimis et al., 2003, 2005; Sanfilippo et al., 2019). Some of the samples have $\Delta\epsilon_{\text{Hf}}$ up to ~ 10 (Fig. 6c), indicating that the ancient component experienced some extents of partial melting in the presence of garnet (resulting in higher Lu/Hf in the residue) (*i.e.* Chauvel and Blichert-Toft, 2001). Nonetheless, the Sr-Nd-Hf isotope compositions of Type 1 pyroxenites are consistent with their derivation from recycled oceanic lithosphere trapped in the convective mantle.

4.2. Contribution of subducted sediments from a *mélange* matrix

The homogeneous major- and trace-element compositions of Type 2 clinopyroxenites are consistent with initially crystallized from, or subsequently reequilibrated with, the melt from which they inherited a geochemically enriched signature. Their convex-upward REE patterns (Fig. 5c), commonly observed in cumulate pyroxenites (*i.e.* McDonough and Frey, 1989; Bizimis et al., 2005; Medaris et al., 2013), thus would reflect the composition of the parental melt. The calculated melt in equilibrium with these pyroxenites (assuming that the reconstructed bulk-rock compositions are representative of the primary cumulate Cpx) is strongly enriched in LREE and LILE and depleted in HFSE (Fig. 7). Mantle-derived magmas with similar trace-element compositions include IAB, carbonatites and potassic/ultrapotassic magmas (*i.e.* melilitites, kimberlites, lamproites). Considering the abundance of CO₂ inclusions and C-bearing apatite in the Type 2 pyroxenites (Fig. 2g; O'Reilly et al., 1990; O'Reilly and Griffin, 2000) and their strongly negative Pb anomalies (Fig. 5d), carbonatites or similarly CO₂-rich hydrous melts are considered as good candidates.

In the pseudoternary system of Fo-CaTs-Qz (Fig. 8), the compositions of Type 2 clinopyroxenites lie on the silica-poor side of the high-pressure thermal divide (En-CaTs join), as for Type 1, but trend markedly into the nepheline-normative field. Such compositional variation has been interpreted elsewhere as due to the presence of minor amounts of silica-poor minerals (Ol, Amp, Spl and/or other oxides) in the source (Keshav et al., 2004). Alternatively, experimental studies suggest that it may result from melt-peridotite interaction involving partial melts derived from eclogitized MORB (Fig. 8; Yaxley and Green, 1998; Mallik and Dasgupta, 2012). This latter alternative is consistent with Type 2 clinopyroxenites having low Cr and Ni contents, like Type 1 and 3 clinopyroxenites,

but relatively high MgO and low Al₂O₃ and Na₂O contents, which resemble those of the Grt websterites (Fig. 4).

Type 2 clinopyroxenites display low ¹⁴³Nd/¹⁴⁴Nd and ¹⁷⁶Hf/¹⁷⁷Hf (Fig. 6), and high ⁸⁷Sr/⁸⁶Sr, even more enriched than EMIL (Zindler and Hart, 1986) and comparable to pelagic sediments. Because of their very low ⁸⁷Rb/⁸⁶Sr (0.003–0.013), these pyroxenites cannot have developed such high ⁸⁷Sr/⁸⁶Sr (0.7142–0.7157) solely by radiogenic ingrowth following crystallization. They also cannot be contaminated by crustal sediments during ascent because they show very different Sr-Nd isotopes from the host basalt; the radiogenic Sr thus reflects the characteristics of their parental melts, which probably incorporated a sedimentary component. This is supported by the negative Eu anomalies and low Sr/Nd ratios of Type 2 pyroxenites (Figs. 5 and 6; Ben Othman et al., 1989). In particular, they plot on the mixing line between Sr-enriched mantle and a bulk-sediment end-member in both Sr-Nd and Nd-Hf space, whereas their trend differs markedly from mixing lines involving sediment-derived melts (Fig. 6b, d). The source of Type 2 pyroxenites thus requires a physical mixing of the subducted components and the ambient peridotite (Nielsen and Marschall, 2017), probably along the slab-mantle interface where high-pressure *mélanges* are formed (Marschall and Schumacher, 2012). It is thus likely that the carbonatitic, or CO₂-riched hydrous parental melts of Type 2 clinopyroxenites originated from a mixed source of peridotite and recycled oceanic crust, where the contribution from carbonated pelagic sediments to melting is significant, and is now reflected in their strong LREE enrichment, HFSE depletion and radiogenic Sr isotopes (Figs. 5 and 6; Ben Othman et al., 1989). Experimental studies suggest that the liberation of carbonated melts from pelagic sediments occurs at higher temperature than that of a modern slab-mantle interface at sub-arc depths (Tsuno and Dasgupta 2011, 2012; Tsuno et al., 2012), thus potentially requiring solid-state advection of sediment piles into the mantle wedge by diapirs or plumes (Currie et al., 2007; Tsuno et al., 2012). We therefore suggest that the enriched signature of Type 2 pyroxenites may be inherited from the melting of a *mélange*-like component, probably incorporated in the deep parts of the convecting mantle wedge.

4.3. From mantle-wedge heterogeneity to lithospheric percolation processes

Lu et al. (2018) showed that the Grt websterites from Lakes Bulenmerri and Gnotuk probably were derived from mantle-wedge peridotite melts. In contrast, we have shown above that, on the one hand, the crystallization of partial melts from a pyroxenite-dominated source has resulted in relatively Mg-poor (Type 1) Grt clinopyroxenites; this source may have contained a large proportion of hydrothermally altered oceanic lithosphere. On the other hand, CO₂-rich melts derived from a *mélange* of altered oceanic crust (including pelagic sediments) and ambient peridotite appear to have produced the relatively MgO-rich (Type 2) Spl-Grt clinopyroxenite segregates.

In this context, it can be envisaged that the compositional variability of Type 3 clinopyroxenitic xenoliths results from the metasomatic overprint of a protolith comparable to Type 1 clinopyroxenites by the carbonatitic melts parental to Type 2 clinopyroxenites. These melts could also potentially be responsible for the metasomatic overprint observed in the websterites and the host peridotites, as suggested for instance by Grt-websterite samples showing trace-element patterns comparable to those of Type 2 clinopyroxenites (Lu et al., 2018). Alternatively, Type 3 clinopyroxenites may represent cumulates from a melt (mixture) that was compositionally intermediate between the parental melts of Type 1 and Type 2 clinopyroxenites. However, the presence of high-Cl Ap locally associated with Amp (which also occurs as distinct veins)

in Type 3 clinopyroxenites (Fig. 2i) clearly indicates modal metasomatism. Moreover, the evolution of REE patterns from depleted, U-shaped (Type 1) to variably enriched LREE (Type 3 clinopyroxenites) suggests that chromatographic re-equilibration accompanied this metasomatic process (Navon and Stolper, 1987). It is thus more likely that Type 1 and Type 3 clinopyroxenites are cogenetic (as suggested by their comparable major-element systematics) and differ only in the extent of their metasomatic overprint.

Type 3 clinopyroxenites have relatively low $^{143}\text{Nd}/^{144}\text{Nd}$ and $^{176}\text{Hf}/^{177}\text{Hf}$, and $^{87}\text{Sr}/^{86}\text{Sr}$ intermediate between Type 1 and Type 2 clinopyroxenites, which are comparable to the websterites (Fig. 6). In the Hf–Nd space, metasomatism of a depleted protolith by a melt in isotopic equilibrium with Type 2 clinopyroxenites, can explain the locus of the websterites and Type 3 clinopyroxenites. However, this process does not fully account for their vertical spread in the Nd–Sr space, which would require Nd to be more readily reequilibrated to the melt composition than Sr, and thus a relatively low Sr/Nd melt.

Alternatively, chromatographic re-equilibration during the percolation of a low-volume fluid or melt in a porous-flow regime, as defined by Navon and Stolper (1987), can also produce isotopic trends that differ from simple mixing models (Le Roux et al., 2009; Tilhac et al., 2020). Chromatography can selectively enrich incompatible elements so that LREE become strongly fractionated from the MREE (i.e. Bedini et al., 1997; Le Roux et al., 2007; Tilhac et al., 2016). This is observed in Type 3 clinopyroxenites, whose $(\text{La}/\text{Nd})_{\text{N}}$ exceeds those of both Type 1 and Type 2 samples (Fig. 6c).

The same mechanism could be responsible for their selective enrichment in LILE. Slow diffusional reequilibration, for instance due to large grain sizes such those observed in the xenoliths studied here (Fig. 2h), may also induce strong local chemical gradients leading to differential reequilibration of radiogenic isotope systems (Tilhac et al., 2020). However, none of the tested combinations of partition coefficients and percolation-diffusion parameters could reproduce jointly the observed variations in Nd–Hf and Nd–Sr spaces (not shown). We thus believe that the main metasomatic melt component for Type 3 clinopyroxenites must be less radiogenic in Sr, and to a lesser extent in Nd and Hf, than Type 2 clinopyroxenites (Fig. 6a, c). Such compositions could be ascribed to basalts in the recycled oceanic lithosphere or mafic veins/dykes in the mantle wedge. The presence of such lithologies is consistent with the Nd-isotope variability of Type 1 pyroxenites, and in turn potentially reflects the heterogeneity of the recycled lithosphere. From a trace-element and isotopic perspective, it is thus reasonable to envisage that a recycled oceanic basaltic melt and/or carbonatitic melts (potentially sediment-derived) parental to Type 2 clinopyroxenites have percolated and metasomatized protoliths similar to those of Type 1 clinopyroxenites to produce Type 3 clinopyroxenites.

4.4. Implications for temporal variations in subsequent volcanism

The xenoliths brought to the surface by NVP basalts at Lakes Bullenmerri and Gnotuk reveal that the lithosphere beneath South-eastern Australia has received melt contributions from a heterogeneous mantle-wedge source (i.e. O'Reilly, 1989 and reference here). We have shown that melt-peridotite interaction probably occurred at lithospheric levels, when carbonatitic melts with an enriched, sedimentary signature infiltrated into the segregates (and their host peridotites) derived from earlier intrusion of silicate melts. The silicate melts may be derived in turn from ambient mantle-wedge peridotite and/or pyroxenite. We envisage below the implications of such scenario and the resulting “lithospheric memory” for the compositional evolution of the related basaltic magmatism (Fig. 9).

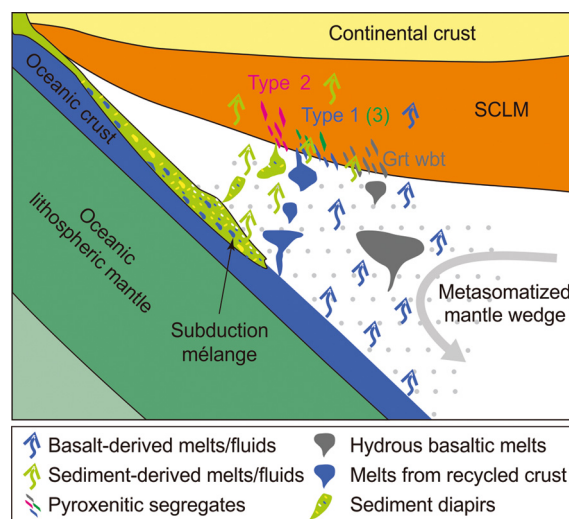


Fig. 9. Schematic illustration of the proposed scenario for the formation of the pyroxenite xenoliths from Lakes Bullenmerri and Gnotuk, SE Australia. Proto-Pacific subduction transported the whole oceanic slab package into the convective mantle wedge, resulting in mantle heterogeneity. The garnet websterites originated from a metasomatic mantle-wedge induced by corner flow (Lu et al., 2018). Type 1 and Type 3 clinopyroxenites were formed as segregates from partial melts of recycled oceanic crust, while Type 2 clinopyroxenites derived from melts of a hybrid mélange-like source with physically mixed carbonatite-bearing pelitic sediments, recycling oceanic crust and peridotite. These melts subsequently migrated upwards and intruded the overlying lithospheric mantle along channels or cracks and metasomatized ambient peridotites and pre-existing pyroxenitic segregates.

Basaltic volcanism is widespread over an area of more than 700 km east to west and ~300 km north to south across the state of Victoria. Basaltic rocks with ages ranging from 95 to 18 Ma, referred to as the “Older Volcanic Province” (OVP), are mainly nepheline-normative basalts. The OVP have Nd–Pb isotopes trending towards EMII, probably indicating that the primary magmas of the OVP were generated by mixing of MORB-source mantle and a low-degree partial melts derived from a carbonatite-metasomatized peridotite (Oostingh et al., 2016). However, some OVP with high Mg# also have low SiO_2 contents and high FeO and $\text{CaO}/\text{Al}_2\text{O}_3$ (Price et al., 2014), which may be markers of pyroxenite in the source (i.e., Lambert et al., 2013). Accordingly, Type 2 pyroxenites in this study show carbonatite-like trace-element patterns and EMII-like isotopic characteristics, and their parent melts may be an appropriate candidate for the source of the OVP basalts. Therefore, we suggest that the source of the OVP may involve the depleted asthenospheric mantle and Type 2 clinopyroxenites.

Volumetrically more abundant (~150,000 km^2) younger lavas gave birth to the “Newer Volcanic Province” (NVP) in western Victoria. Magmatic activity began at 4.6 Ma, reached a peak at 2.6 Ma, and continued until the most recent eruption at about 4.6 ka. The NVP magmatism was subdivided into two series on a geomorphological and compositional basis: the early Plains and slightly later Cones (Wellman and McDougall, 1974). The NVP Cones dominantly comprise silica-undersaturated alkali basalts with OIB-type trace element characteristics and host abundant xenoliths. They show relatively limited Sr–Nd–Pb isotopic variations and evolve towards the HIMU mantle end-member, probably generated by mixing sources similar to the DMM and peridotites metasomatized by alkaline magmas (Oostingh et al., 2016).

Experimental studies suggest that the alkalic OIBs with HIMU signature can be generated by reaction/mixing between a carbonated silicate melt in equilibrium with MORB-like pyroxenite and peridotites/peridotitic melts. (Gerbode and Dasgupta, 2010). The carbonated MORB-like pyroxenite could be the source of Type 1 clinopyroxenites (see discussion above). In addition, Cpx–Grt pairs

in Type 1 clinopyroxenites gave Sm-Nd isochron ages of 56 Ma, predating the NVP volcanics. This age should represent a cooling age, which is younger than its actual formation ages, because pervasive recrystallization can easily reset the Sm-Nd system and even Lu-Hf system. Therefore, it is reasonable that Type 1 clinopyroxenites may have been involved in the source of the NVP. Furthermore, considering that the peridotites have trace-element patterns and isotopic compositions similar to the garnet websterites in this area, we propose that the alkaline magmas may be similar to the basaltic magmas from which garnet websterites crystallized (Griffin et al., 1988; Stolz and Davies, 1988). Therefore, the source heterogeneity of the OVP and NVP may be related to the distribution of dikes/veins of the different types of pyroxenites, in turn reflecting previous magmatic episodes involving components of different origins (e.g., recycled basalts and sediments, ambient mantle wedge).

The identification of the geochemical signature of such subduction-related components in SEA volcanics may thus simply reflect the contribution of lithospheric lithologies (pyroxenites in particular) carrying such signatures, rather than a role for a subduction system in the generation of the Tertiary basalts (i.e., O'Reilly and Griffin, 1988; Griffin et al., 1988). In this regard, the magmatic/metamorphic history recorded by the mantle beneath SEA constitutes a "lithospheric memory", potentially long-lived, and only revealed during later tectonic reactivation of the region (in this case, the Cenozoic rifting; O'Reilly and Zhang, 1995). From a geodynamic perspective it could be envisaged, for instance, that the presence of pyroxenite fragments in the source of the SEA basalts is related to edge-driven convection (EDC) shearing the irregular base of the lithosphere as the Australian Plate moved northward (i.e. Lister and Etheridge, 1989). While the compositional variability of both erupted basalts and their associated xenoliths reflects the presence of a heterogeneous mantle-wedge beneath SEA, the latter was probably not involved in the source of these rift-related basalts. As a corollary, the volume of subduction-related magmatism through time may be overestimated if it is only based on the geochemistry of erupted lavas. Studies utilizing the complementarity of lavas/xenoliths are likely to be more realistic, particularly where tectonic reconstructions are concerned.

5. Conclusions

Different types of Grt clinopyroxenites were recognized in the mantle xenoliths from Lakes Bullenmerri and Gnotuk, Southeastern Australia based on their modal, major-, trace-element, and Sr-, Nd- and Hf-isotope compositions. Compared to Grt websterites from the same locality, the clinopyroxenites are characterized by low MgO, low Cr₂O₃ and Ni contents, and are interpreted as segregates from melts derived from recycled oceanic lithosphere with variable proportions of sedimentary components. They also exhibit a striking correlation between crystallization indices and the contents of highly incompatible elements such as the LREE, Sr and Th, ascribed to metasomatic overprint by carbonatitic melts carrying the sediment signature, in a good agreement with their Sr-Nd-Hf isotope systematics. The garnet pyroxenites thus record the impact of various slab-derived melts on the chemical heterogeneity of the mantle wedge. We propose that this subduction signature was subsequently stored in the lithosphere (where the pyroxenitic melts intruded), until its tectonic reactivation during Cenozoic rifting. As a consequence, the identification of various subduction components in the source of the rift-related lavas of the Southeastern Australia magmatic provinces may simply reflect the presence and contribution of pyroxenite fragments in their source, a "lithospheric memory" of subduction, rather than the reflection of contemporaneous subduction tectonics. This is thus a rare example where the intermediate and final products of the subduction

factory can be observed in a single, tectonically well-constrained region, allowing for a direct assessment of the potential role of the lithospheric filter.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epsl.2020.116365>.

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